JIANGYIFEI ZHU

Carnegie Mellon University ◦ CIC, 4720 Forbes Avenue, Pittsburgh, PA, USA, 15213

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Ph.D. in Electrical and Computer Engineering — May 2028
Advisors: Prof. Justin Chan & Prof. Swarun Kumar
Research Areas: Signal Processing, ML for sensor data, Wireless Sensing, Healthcare
Winner of the Qualcomm Innovation Fellowship (6% acceptance rate)

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering, GPA: 3.9/4.0 — May 2024
Selected Coursework: Wearable Health Technologies, Computer Network, Real-time Embedded System

University of Massachusetts Amherst — Amherst, MA
Bachelor of Science in Computer Science, GPA: 3.8/4.0, **Dean's List** — February 2022
Bay State Scholar (A scholarship worth 50% of tuition)

Lewis & Clark College — Portland, OR
Coursework Completed Toward Bachelor's Degree — September 2017 - May 2019

PUBLICATIONS

- Jiangyifei Zhu*, Yuzhe Wang*, Tao Qiang*, Vu Phan, Zhixiong Li, Evy Meinders, Eni Halilaj, Justin Chan, Swarun Kumar, "**GigaFlex: Contactless Monitoring of Muscle Vibrations During Exercise with a Chaos-Inspired Radar**", *ACM SenSys 2026*
- Jiangyifei Zhu*, Kuang Yuan*, Akarsh Prabhakara, Yunzhi Li, Gongwei Wang, Kelly Michaelsen, Justin Chan, Swarun Kumar, "**Measuring multi-site pulse transit time with an AI-enabled mmWave radar**", *Nature Communications*
- Siqi Zhang, Xiyuxing Zhang, Duc Vu, Tao Qiang, Clara Palacios, Jiangyifei Zhu, Yuntao Wang, Mayank Goel, Justin Chan, "**LubDubDecoder: Bringing Micro-Mechanical Cardiac Monitoring to Hearables**", *ACM CHI 2026*
- Yawen Liu, Akarsh Prabhakara, Jiangyifei Zhu, Shenyi Qiao, Swarun Kumar, **"Shape-programming Robotic Reflectors for Wireless Networks"**, *IEEE ICRA 2025*

EXPERIENCE

Bosch Research — Pittsburgh, PA
Multi-Modal Cyber-Physical Sensing Intern — June 2026 - Present
Skills: mmWave Radar, Multimodal Sensing, Distributed Systems, Agentic Workflows, Large Data Collection

- Developed three time-synchronized multimodal sensing rigs supporting day-long, multi-view data collection; used a modular mechanical design with a network stack on each rig to enhance deployment.
- Built real-time mmWave radar demos for 3D object localization and human pose tracking; enabled streaming of perception outputs to devices on a shared mesh network.
- Developed a documentation-driven agentic deployment workflow that enables AI agents to configure, troubleshoot, and replicate distributed sensing systems.
- Manage a multimodal data collection campaign targeting 50+ hours of recordings.

Radar Necklace for Swallow Sensing — Pittsburgh, PA
Advised by Justin Chan and Swarun Kumar — January 2026 - Present
Skills: Radar/IMU Sensing, Cross-Modal Representation Distillation, Deep Learning, Wearable Systems

- Contributed to the design of a necklace-sized integrated circuit board with wireless data transmission.
- Developed an IMU-to-radar representation distillation framework for radar-only swallow detection, achieving a 16% improvement in ROC-AUC over supervised CNN training.

Remote Muscle Monitor via Radar Pittsburgh, PA

Advised by Justin Chan and Swarun Kumar May 2025 - November 2025

Skills: mmWave Radar, Chaos-Based Signal Processing, Deep Learning

- Built an FMCW mmWave pipeline from raw IF signals to Range-Doppler and Range-Angle maps; applied chaos-based phase analysis to isolate 1–100 Hz muscle vibrations from gross body motion.
- Designed a dual-branch neural network with multi-scale temporal features and SE attention to fuse subject-specific embeddings, achieving ~6% vibration-modeling error.
- Developed a logistic-regression fatigue classifier using phase-space features, achieving 0.86 AUC for high-fatigue states (1 repetition in reserve) versus standard repetitions.

Contactless Multi-Site Pulse Sensing by Radar Pittsburgh, PA

Advised by Justin Chan and Swarun Kumar September 2023 - March 2025

Skills: Radar Signal Processing, Adaptive Filtering, Deep Learning

- Implemented Range FFT and beamforming on an 86-virtual-antenna mmWave array (1.4° angular resolution) to isolate ~30 μm physiological motion from respiratory interference.
- Developed autocorrelation-based spatial-bin ranking to extract pulse signals from the heart, neck, and wrist despite multipath interference and clothing occlusion.
- Co-developed a multi-branch network with bidirectional LSTMs and cross-region attention for pulse transit time and diastolic blood pressure estimation (PTT R=0.89; DBP R=0.92).

Software Engineer @ Huawei Digital Technologies Co., Ltd Beijing, China

Cluster Computer Solution Team March 2022 - August 2022

Skills: PyTorch, Hardware Acceleration, System Optimization

- Optimized PyTorch GNN operators for Ascend NPUs, Kunpeng ARM, and Intel x86; automated benchmarking saved ~4 hours per test cycle.
- Designed cluster resource pre-partitioning that reduced environment deployment time by over 95%, from 30 minutes to under 1 minute.

SKILLS

Programming: Python, MATLAB, C/C++, Java, Dart/Flutter, JavaScript, C#
ML & Signal Processing: Deep Learning (PyTorch/TensorFlow), Time-series & Chaos features, Radar/IMU/sEMG/PPG processing, Adaptive filtering & Periodicity analysis
Software & Tools: Linux, Git, Docker, LaTeX, mobile & Web development
Sensing/Hardware: TI mmWave Radar (AWR series), PPG, IMU, ZED X, Embedded Systems, Arduino, DAQ Systems.

SCHOLARSHIPS

- Liang Ji-Dian Fellowship at Carnegie Mellon University Feb 2026
- Qualcomm Innovation Fellowship Winner May 2025
  Selected for the proposal *"Contactless Cardiovascular Health Monitoring using AI-enabled mmWave Radars"*
- Center for Machine Learning and Health Fellowship at Carnegie Mellon University April 2025

PROFESSIONAL SERVICE

Reviewer: ACM UIST 2026; ACM CHI 2026; IEEE ICC 2026 (IoT Symposium); ACM IMWUT 2025, 2026

TEACHING, TALKS & OUTREACH

Guest Lecture, CMU 18-848: Introduction to FMCW Radar. September 2026
Guest Lecture, CMU 18-213/18-613: Contactless Sensing via Radar. April 2026
Guest Lecture, CMU 18-441/18-741: Advanced Topics in Wireless Technology. November 2024, 2025
Guest Lecture, CMU 17-320/17-720: Wireless Sensing and Health. September 2025
Research Talks: ACM SenSys (GigaFlex); UPMC Digital Health Summit (Cardiovascular sensing). 2026
Museum Demo, Kamin (Carnegie) Science Center: contactless breathing and heart-rate sensing. May 2026
K–12 Outreach, CMU Gelfand Center: "Detect Like a Superhero: Using Waves to Sense!" June 2025